Name _____ Period _____ Date _____

4-4 Lesson Quiz: Adding and Subtracting Rational Expressions

short form: questions 1, 4 and 5 of the Savvas 4-4 Lesson Quiz. Show your work for every question.

1. Add $\frac{5}{2x+1} + \frac{8}{4x+2}$, with $x \neq -\frac{1}{2}$.

Ⓐ $\frac{9}{2x+1}$

Ⓑ $\frac{13}{4x+2}$

Ⓒ $\frac{5x+8}{4x+2}$

Ⓓ $\frac{18}{2x+1}$

Work

4. Simplify $\frac{\frac{1}{x} - \frac{1}{x+2}}{2}$.

Ⓐ $\frac{1}{x+2}$; $x \neq -2$

Ⓑ $\frac{1}{x}$; $x \neq 0$

Ⓒ $\frac{1}{x(x+2)}$; $x \neq -2$ or 0

Ⓓ $\frac{1}{x(x-2)}$; $x \neq 2$ or 0

Work

5. The equation $r = \frac{1}{\frac{1}{r_1} + \frac{1}{r_2}}$ represents the total resistance, r, when two resistors whose resistances are r_1 and r_2 are connected in parallel. Find the total resistance when r_1 is x and r_2 is $x + 1$.

Ⓐ $\frac{1}{2x+1}$; $x \neq -1$, $-\frac{1}{2}$, or 0

Ⓑ $2x + 1$; $x \neq -1$ or 0

Ⓒ $\frac{x(x+1)}{2x+1}$; $x \neq -1$, $-\frac{1}{2}$, or 0

Ⓓ $\frac{2x+1}{x(x+1)}$; $x \neq -1$ or 0

Work

Scoring: Questions 1 and 4 are 10 points each; Question 5 is a bonus of up to 5. Total out of 20.

4-4 Lesson Quiz, short form: answer key

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1. A, $\frac{9}{2x+1}$ $(x \neq -\frac{1}{2})$. Factor: $4x+2 = 2(2x+1)$, so the LCD is $2(2x+1)$: $\frac{5\cdot 2}{2(2x+1)} + \frac{8}{2(2x+1)} = \frac{18}{2(2x+1)} = \frac{9}{2x+1}$. Distractor B ($\frac{13}{4x+2}$) adds the numerators without rewriting $\frac{5}{2x+1}$ over $4x+2$. D ($\frac{18}{2x+1}$) is the right numerator over the wrong denominator: the factor 2 was dropped instead of divided out. C ($\frac{5x+8}{4x+2}$) is not equal to the sum (try $x = 0$: the sum is 9, C gives 4). Savvas quiz item 1, DOK 1, skill A86; taught by Example 3. **10 points:** 3 for the LCD $2(2x+1)$, 4 for the numerator 18, 3 for choice A (divided out).

4. C, $\frac{1}{x(x+2)}$**;** $x \neq -2$ **or** 0**.** Multiply top and bottom by the LCD $x(x+2)$: $\frac{(x+2)-x}{2x(x+2)} = \frac{2}{2x(x+2)} = \frac{1}{x(x+2)}$. Distractor D has the wrong sign in the factor $x-2$. Savvas quiz item 4, DOK 2, skill A86; taught by Example 6 and Try It 6. **10 points:** 3 for the LCD $x(x+2)$, 4 for clearing the small fractions to $\frac{2}{2x(x+2)}$, 3 for choice C.

5. C, $\frac{x(x+1)}{2x+1}$**;** $x \neq -1, -\frac{1}{2}$**, or** 0**.** Substitute: $r = \frac{1}{\frac{1}{x}+\frac{1}{x+1}}$. Multiply top and bottom by $x(x+1)$: $\frac{x(x+1)}{(x+1)+x} = \frac{x(x+1)}{2x+1}$. $x = -\frac{1}{2}$ is excluded because it makes $\frac{1}{x} + \frac{1}{x+1} = 0$, so r is undefined there. Distractor D is the reciprocal (the sum $\frac{1}{x} + \frac{1}{x+1}$ before dividing). Savvas quiz item 5, DOK 2, skill A86; Example 6 (the packet's Practice block is its parallel). **Bonus 5:** 2 for the substitution, 3 for choice C with work.

Items left off this form (available for a retake or Form B): Savvas item 2, LCM of $(x+3)^2(x-2)$ and $(x+3)(x^2-16)$ (answer D, $(x+3)^2(x-2)(x^2-16)$; DOK 2, A86; Example 2 content, the LCD step of Example 3); Savvas item 3, subtract $\frac{3}{x-1} - \frac{4}{x+3}$ (answer A, $\frac{-x+13}{(x-1)(x+3)}$, $x \neq -3$ or 1; DOK 1, A86; Example 4 content, cut: Example 3's steps with the -1 distributed, practised in IXL FEX).

IXL replacement: SmartScore 70 or above on FEX and YD8 (the starred skills; YD8 also feeds the item 5 bonus) replaces this quiz with 20/20.